

Classical-coordinate interval certificates

PRIME.RDAGGER.CLASSICAL_CERT.01 — round 471

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Abstract

Round 469 redirected resources to fixed support length. This note executes foundation F2: a Guinand–Weil quadratic form is assembled directly in the classical coordinate, dictionary-free (no fold, no selected window, no $R^\dagger$). Validated Cholesky on the interval Toeplitz matrix of an explicit finite-dimensional grid family $F_{L,\delta}$ yields

$$Q_L \succeq 0 \quad \text{on } F_{L,\delta}$$

at the calibration point $L = 0.8$ and then up through $L_{16} = 4 \log 2 = 2.772588\dots$. The type is **GRID_CERTIFIED**, never $\lambda_*(L) \geq 0$. The remaining gap is named F2b. No statement about infinitely many k . No RH claim.

Status. GRID_CERTIFIED($L_{\max} = 2.7726$) on the sealed meshes below. No promotion of an RH claim. No claim for or against RH.

Claim boundary. Every positivity statement in this note is of type **GRID_CERTIFIED**(L, δ): the quadratic form of one explicit finite-dimensional family of piecewise-constant test functions is positive semidefinite. This is strictly weaker than $\lambda_*(L) \geq 0$, which infimizes over all L^2 functions supported in $[-L, L]$. Closing that gap is F2b, not this round. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Classical object (r461, taken literally)

Write $Q_W(h)$ for the Guinand–Weil functional of the even autocorrelation $g = h * \tilde{h}$. After r461,

$$\lambda_*(L) = \inf_{\substack{0 \neq h \in L^2(\mathbb{R}) \\ \text{supp } h \subset [-L, L]}} \frac{Q_W(h)}{\|h\|_2^2}, \quad L_k = \frac{k \log 2}{4}.$$

Then $\text{supp } g \subset [-2L, 2L]$. The classical Yoshida/Bombieri zone is the prime-blind range $U < \log 2$, i.e. $L < (\log 2)/2 \approx 0.34657$. The external preprint arXiv:2608.24827 certifies a strictly positive floor for *all* even L^2 test functions with $\text{supp } h \subset [-0.8, 0.8]$. The translation “ $R^\dagger(W_k) \succeq \frac{1}{2}I \implies \lambda_*(L_k) \geq 0$ ” remains conditional on the named r462/r463 bridges; this round does not use that dictionary.

2 Self-defined form

For even compactly supported g ,

$$Q_W(g) = A(g) - P(g) + \Pi(g),$$

$$\begin{aligned}
P(g) &= \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} g(\log n), \\
\Pi(g) &= 4 \int_0^\infty g(x) \cosh(x/2) dx, \\
A(g) &= -(\gamma + \log \pi) g(0) + 2 \int_0^\infty \frac{g(0)e^{-2w} - g(w)e^{-w/2}}{1 - e^{-2w}} dw.
\end{aligned}$$

Compact support of g cuts P to the finite range $n \leq e^{2L}$. (The briefing’s $n \leq e^L$ mixes the autocorrelation length $U = 2L$ with L ; the probe follows r461.) On piecewise-linear g the pole term is closed-form. The archimedean integral is enclosed by a Bernoulli expansion of $x/(1 - e^{-x})$ on a prefix $[0, \eta]$ and a geometric Bose series with an integral tail bound on $[\eta, 2L]$.

3 Grid family $F_{L,\delta}$

Let h be piecewise constant on $n = 2L/\delta$ equal bins of $[-L, L]$. Then g is a linear combination of tents of half-width δ , and Q_W becomes an $n \times n$ Toeplitz form Q_L in the bin coefficients. Interval entries are sealed: P from an integer Eratosthenes list of prime powers, Π exact, A as above. A Higham-validated Cholesky factor of the midpoint, with the interval-radius row-sum subtracted, produces a rigorous lower bound μ on $\lambda_{\min}(Q_L)$. If $\mu > 0$ the type is

$$\text{GRID_CERTIFIED}(L, \delta) \iff Q_L \succeq 0 \text{ on } F_{L,\delta}.$$

This is scramble-sensitive: P evaluates g at the literal nodes $\log n$. A source-position scramble moves those nodes and changes Q_L . The bound is not a $q^\dagger$ pairing and is not scramble-invariant (r469 anti-list item 3).

4 Sealed table

Probe `classical_cert_probe.py`, SPEC_SHA 18eb2a33651585f5. Smoke 14/14 in 0.19s; full 25/25 in 2.3s. All interval radii are $\sim 2 \cdot 10^{-11}$. Prime lists are pinned by $(n_{\max}, \#\{p^k\})$.

L	n	δ	type	external comparison
0.30	8, 16	0.075, 0.0375	GRID_CERTIFIED	inside Yoshida/Bombieri
0.80	8, 16, 24	0.20–0.067	GRID_CERTIFIED	preprint zone (grid subclass)
$L_5 = 0.866434$	8, 16	0.217, 0.108	GRID_CERTIFIED	beyond $L = 0.8$
1.0	20	0.10	GRID_CERTIFIED	—
1.2	24	0.10	GRID_CERTIFIED	—
$L_8 = 1.386294$	28	0.099	GRID_CERTIFIED	—
1.4	28	0.10	GRID_CERTIFIED	—
1.6	32	0.10	GRID_CERTIFIED	—
1.8	32	0.1125	GRID_CERTIFIED	—
2.0	32	0.125	GRID_CERTIFIED	—
$L_{12} = 2.079442$	32	0.130	GRID_CERTIFIED	—
$L_{16} = 2.772589$	36	0.154	GRID_CERTIFIED	r465/r468 top k

The kill “calibration at $L = 0.8$ must be green” holds at three meshes. No row is typed as $\lambda_*(L) \geq 0$. No row is **OPEN** on the sealed schedule.

5 Discretization gap (F2b)

The infimum $\lambda_*(L)$ ranges over a Sobolev/Weil test class on which g is continuous, so that $P(g)$ is well-defined pointwise and $A(g)$ is the classical Mellin–digamma pairing. Piecewise-constant h are

dense in $L^2([-L, L])$, but L^2 is the wrong topology for Q_W : a jump in h makes g only Lipschitz, and the Weil class is usually C_c^2 (or the Fourier transform of a nonnegative even function of compact support). The missing lemma is therefore exactly F1(iii) of the r469 judgment, restricted to one fixed L :

- dyadic (or grid) autocorrelations are dense in the Weil test class at support length L ;
- Q_W is continuous there, with an explicit modulus $\omega_L(\delta)$.

If a mesh with bin width δ satisfies $\mu(Q_L) > \omega_L(\delta)$, the grid certificate upgrades to $\lambda_*(L) \geq \mu - \omega_L(\delta)$. At $L = 0.8$, $n = 24$, one has $\delta = 1/15$ and $\mu \approx 1.32 \cdot 10^{-4}$. A C^2 modulus of order $O(\delta^2)$ would close at a plausible constant; an $O(\delta)$ modulus would need a constant $\lesssim 2 \cdot 10^{-3}$. Neither bound is proved here. F2b is standard analysis at fixed L and does not restate a cofinal quantifier.

6 Kill table

Gate	Result
Calibration $L = 0.8$ green	pass ($n = 8, 16, 24$)
Yoshida/Bombieri $L = 0.30$ green	pass ($n = 8, 16$)
Interval enclosures sealed	pass (radius $\sim 2 \cdot 10^{-11}$)
Prime term exact, $n \leq e^{2L}$ pinned	pass
Determinism (two full runs)	pass (byte-identical PIN dump)
Grid-vs-full typing in every statement	pass (GRID_CERTIFIED only)
Scramble gate declared before the bound	pass (nodes $\log n$)
No ∞ -many- k statement	pass
Zero-oracle firewall	pass

7 What this does not do

It does not close F1(i) or F1(ii). It does not prove F2b. It does not convert any $R^\dagger$ certificate into $\lambda_*(L_k) \geq 0$. It does not claim positivity of Q_W on the whole Weil class at any L , including $L = 0.8$. The preprint theorem at $L = 0.8$ remains the external full-class certificate; the grid calculation is an independent calibration on a subclass. No RH claim.